

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and infrastructure Engineering) SEMESTER - 4 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTCIC405 - HIGHWAY ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 The surveys of highway alignment are completed in how many stages? a) 1 b) 2 c) 3 d) 4
- 2 The current highway development works in India are undertaken by? a) NHAI b) State Government c) Govt of India d) NHDP
- 3 The shape of the camber, best suited for cement concrete pavements, is a) straight line b) parabolic c) elliptical d) combination of straight and parabolic
- 4 Coefficient of friction is less when the pavement surface is a) rough b) dry c) smooth and dry d) smooth and wet
- 5 If the stopping distance is 60 meters, then the minimum stopping sight distance for two lane, two way traffic is a) 30m b) 60m c) 120m d) 180m
- 6 The transition curve used in the horizontal alignment of highways as per IRC recommendations is a) spiral b) lemniscate c) cubic parabola d) any of the above
- 7 The resistance to flow is measured by _____. a) Flash & Fire b) Viscosity c) Ductility Test d) Penetration Test
- 8 Which of the following is known as design capacity? a) basic capacity b) theoretical capacity c) possible capacity d) practical capacity
- 9 Equivalent factor of passenger car unit (PCU) for a passenger car as per IRC is a) 1.0 b) 2.0 c) 0.5 d) 10
- 10 "Dead Slow" is a a) regulatory sign b) warning sign c) informative sign d) none of the above
- 11 Los Angeles testing machine is used to conduct a) abrasion test b) impact test c) attrition test d) crushing strength test
- 12 Which of the following pavement has greater life? a) Bituminous Road b) Concrete Roads c) Gravel Roads d) Earth Roads

Q2. Solve the following.

- A) Explain the Stopping Sight Distance and Overtaking Sight Distance with a neat sketch. 6
- B) Explain the features of Jayakar Committee 6

- Q3. Solve the following.
- A) Calculate the design speed of a vehicle on a horizontal curve having radius of 100m. with permissible super elevation of 7%. Consider co efficient of friction 0.8. 6
 - B) Draw a neat sketch of std. C/s of National Highway in embankment. 6
- Q4. Solve Any Two of the following.
- A) Write the procedure of construction of cement concrete pavement showing its components. 6
 - B) Explain any four causes of failure of rigid pavement 6
 - C) Write the procedure of construction of WBM road with neat sketch showing various components 6
- Q5. Solve Any Two of the following.
- A) What are the different types of tests carried out on Bitumen in the Laboratory? Explain the Penetration Test in detail. 6
 - B) Explain the Origin and Destination Study. What are the various uses of O & D studies. 6
 - C) Explain the Divisional Island with neat sketch 6
- Q6. Solve Any Two of the following.
- A) Define Intelligent Transport System (ITS) and give some examples of Intelligent Transport System (ITS) in India. 6
 - B) Enlist different types of traffic island and explain any one in brief with neat sketch. 6
 - C) State the types and causes of landslides. 6

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